Problem 9: Polynomial Calculator

TASK

A maths tool stores each polynomial as a singly linked list of terms (coefficient, exponent) in decreasing order of exponent. Two polynomials must be added into a new list in the same order; terms whose coefficients add up to 0 must not appear.

Implement `Node addPolynomials(Node p, Node q)` and print the result in standard form.

CONSTRAINTS

- 0 <= number of terms in each polynomial <= 1000
- Exponents are distinct within one polynomial and are non-negative
- Each input list is visited once

EXAMPLES

Sample 1

```
p = 5x^3 + 4x^2 + 2
q = 5x^2 - 3x + 1

Expected Output: 5x^3 + 9x^2 - 3x + 3
```

Sample 2

```
p = 3x^2 + 1
q = -3x^2 + 4x

Expected Output: 4x + 1
```

COMPLEXITY REQUIREMENTS

State the time and space complexity in terms of the numbers of terms m and n.

HINT

> *This is a merge of two sorted lists: compare exponents, copy the larger one, and add coefficients when the exponents are equal.*
